UNIT - 4

Multi Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Lagrangian method for formulation of equation of motion, influence co-efficient method, lumped mass and distributed mass systems, Stodola method, Holzer's method, modal analysis of free and forced vibrations.

1. Multi-Degree-of-Freedom (MDOF) Systems

System needing n independent co-ordinates. Equations of motion are n coupled 2nd-order ODEs - written in matrix form:

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F(t)\} \quad (1)$$

- $[M]$ = mass matrix, $[K]$ = stiffness matrix, $[C]$ = damping matrix, all $n \times n$, symmetric. $\{x\}$, $\{F\}$ are $n \times 1$ vectors.
- Free undamped vibration: assume $\{x\} = \{X\}\cos(\omega t - \varphi) \rightarrow [K]\{X\} = \omega^2[M]\{X\}$ (eigenvalue problem).
- Non-trivial solution: $|[K] - \omega^2[M]| = 0 \rightarrow$ polynomial of degree n in $\omega^2 \rightarrow n$ natural frequencies $\omega_1 \le \omega_2 \le \ldots \le \omega_n$, and n mode shapes $\{X\}^{(i)}$.
- Exact solution for $n > 3$ by hand is impractical → approximate methods (Rayleigh, Stodola, Holzer, Dunkerley) or computer.

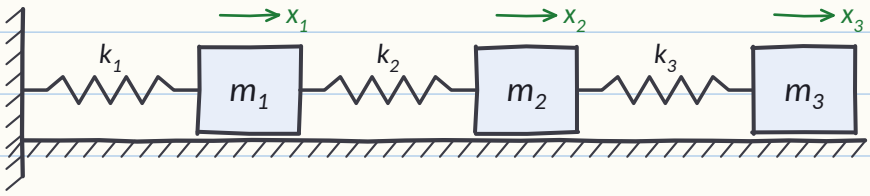


Fig 4.1 Three-DOF chain

2. Lagrangian Method for Equations of Motion

Energy method suited to complex systems (no free-body diagrams, no constraint forces). Uses generalised co-ordinates q_i (independent co-ordinates, number = DOF; any units - length, angle).

Lagrange's equation for $i = 1 \dots n$:

$$d/dt\,(\partial T/\partial \dot{q}_i) - \partial T/\partial q_i + \partial D/\partial \dot{q}_i + \partial U/\partial q_i = Q_i \quad (2)$$

- T = kinetic energy, U = potential energy, D = Rayleigh dissipation function = $\tfrac{1}{2}\sum c_{ij}\dot{q}_i\dot{q}_j$ ($\tfrac{1}{2} c \dot{x}^2$ for dashpot), Q_i = generalised non-conservative force.
- Quadratic forms: $T = \tfrac{1}{2}\{\dot{q}\}^T[M]\{\dot{q}\}$, $U = \tfrac{1}{2}\{q\}^T[K]\{q\}$ → coefficients give $[M]$ and $[K]$ directly: $m_{ij} = \partial^2 T/\partial\dot{q}_i\partial\dot{q}_j$, $k_{ij} = \partial^2 U/\partial q_i \partial q_j$.

Procedure

1. Select generalised co-ordinates $q_1 \dots q_n$.
2. Write T, U (and D, Q_i) in terms of q and $\dot{q}$.
3. Differentiate as per (2) for each q_i → n equations.
4. Arrange in matrix form $[M]\{\ddot{q}\} + [C]\{\dot{q}\} + [K]\{q\} = \{Q\}$.

Example - three-mass chain (Fig 4.1, equal m and k):

$T = \tfrac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2)$, $U = \tfrac{1}{2}k[\,x_1^2 + (x_2 - x_1)^2 + (x_3 - x_2)^2\,]$

$\partial U/\partial x_1 = k(2x_1 - x_2)$; $\partial U/\partial x_2 = k(-x_1 + 2x_2 - x_3)$; $\partial U/\partial x_3 = k(-x_2 + x_3)$

$$m\,[\,1\,0\,0\,;\,0\,1\,0\,;\,0\,0\,1\,]\,\{\ddot{x}\} + k\,[\,2\,{-1}\,0\,;\,{-1}\,2\,{-1}\,;\,0\,{-1}\,1\,]\,\{x\} = \{0\} \quad (3)$$

Example 2 - spring-pendulum (mass m on spring k, which swings): use r (spring length) and θ: $T = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$, $U = \tfrac{1}{2}k(r - l_0)^2 - m\,g\,r\cos\theta$. Lagrange gives the two coupled non-linear equations $m\,\ddot{r} - m\,r\dot{\theta}^2 + k(r - l_0) - m\,g\cos\theta = 0$, $m\,r^2\ddot{\theta} + 2\,m\,r\,\dot{r}\,\dot{\theta} + m\,g\,r\sin\theta = 0$.

Advantage:

Lagrange's method: scalar energies only → easier than Newton for MDOF/constrained systems; constraint forces never appear.

Problem 1a (double pendulum by Lagrange)

Masses m_1, m_2; rods l_1, l_2; co-ordinates θ_1, θ_2 (from vertical).

$T = \tfrac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \tfrac{1}{2}m_2 l_2^2\dot{\theta}_2^2 + m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)$

$U = -(m_1 + m_2)\,g\,l_1\cos\theta_1 - m_2\,g\,l_2\cos\theta_2$

Small angles ($\cos \approx 1$, $\sin \approx \theta$, drop 2nd-order terms) - Lagrange gives:

$$\begin{aligned}(m_1+m_2)\,l_1^2\,\ddot{\theta}_1 + m_2\,l_1 l_2\,\ddot{\theta}_2 + (m_1+m_2)\,g\,l_1\,\theta_1 &= 0\\ m_2\,l_1 l_2\,\ddot{\theta}_1 + m_2\,l_2^2\,\ddot{\theta}_2 + m_2\,g\,l_2\,\theta_2 &= 0\end{aligned} \quad (3a)$$

For $m_1 = m_2$, $l_1 = l_2$ these reduce to the double-pendulum equations of Unit III: $\omega^2 = (2 - \sqrt{2})g/l$ and $(2 + \sqrt{2})g/l$.

3. Influence Coefficient Method

Describes the elastic properties of the system directly from static behaviour. Two forms:

(a) Stiffness influence coefficients k_{ij}

k_{ij} = force required at point i to produce unit displacement at j, with all other points held at zero displacement. Force-displacement: $\{F\} = [K]\{x\}$

$$F_i = \Sigma_j\, k_{ij}\, x_j \quad [K] = [k_{ij}] \text{ (symmetric: } k_{ij} = k_{ji}) \tag{4}$$

Three-mass chain: give $x_1 = 1$, $x_2 = x_3 = 0 \rightarrow$ forces needed: $k_{11} = 2k$, $k_{21} = -k$, $k_{31} = 0$ ... $\rightarrow$ matrix of eqn (3).

(b) Flexibility influence coefficients a_{ij}

a_{ij} = deflection at i due to a unit load at j (no other loads). $\{x\} = [a]\{F\}$

$$x_i = \Sigma_j\, a_{ij}\, F_j \quad [a] = [K]^{-1},\ a_{ij} = a_{ji} \text{ (Maxwell's reciprocal theorem)} \tag{5}$$

For free vibration the force on mass j is the inertia force $-m_j\ddot{x}_j$:

$$\{x\} = -[a][M]\{\ddot{x}\} \rightarrow [a][M]\{X\} = (1/\omega^2)\{X\} \text{ i.e. } [D]\{X\} = \lambda\{X\},\ \lambda = 1/\omega^2$$
$$[D] = [a][M] = [K]^{-1}[M] \text{ (dynamic matrix)} \tag{6}$$

- Flexibility form is preferred when beams/shafts with several masses are given (deflections easy to get from beam formulae); it is the starting point of Stodola's method.

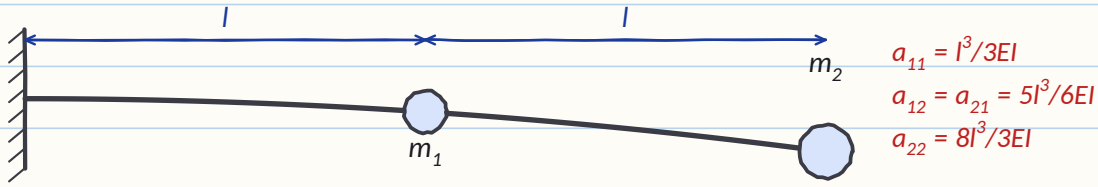


Fig 4.2 Cantilever with two masses - flexibility coefficients

Deflection of cantilever at x due to load P at a ($x \geq a$): $y = P\,a^2(3x - a)/6EI$. With $a = l$: $a_{11} = l^3/3EI$; $x = 2l$: $a_{21} = 5l^3/6EI$; $a = x = 2l$: $a_{22} = 8l^3/3EI$.

Problem 1 (two masses on cantilever)

Massless cantilever (EI constant), $m_1 = m_2 = m$ at l and 2l. Find ω_1, ω_2.

$[a][M] = (m\,l^3/6EI)\ [\ 2\ 5\ ;\ 5\ 16\ ]$. Let $\lambda = 1/\omega^2$, $\mu_0 = m\,l^3/6EI$: $|\ 2\mu_0 - \lambda\ ,\ 5\mu_0\ ;\ 5\mu_0\ ,\ 16\mu_0 - \lambda\ | = 0$

$\lambda^2 - 18\mu_0\lambda + 7\mu_0^2 = 0 \rightarrow \lambda = \mu_0(9 \pm \sqrt{74}) = 17.60\mu_0,\ 0.3977\mu_0$

$$\omega_1^2 = 1/17.60\mu_0 = 0.3409\ EI / (m\ l^3) \rightarrow \omega_1 = 0.584\ \sqrt{(EI/ml^3)}$$
$$\omega_2^2 = 1/0.3977\mu_0 = 15.09\ EI / (m\ l^3) \rightarrow \omega_2 = 3.885\ \sqrt{(EI/ml^3)} \quad (7)$$

Check: $6/17.60 = 0.3409$; $6/0.3977 = 15.09$. Mode 1 ratio $X_2/X_1 = (\lambda - 2\mu_0)/5\mu_0 = 3.12$.

Problem 1b (simply supported beam, two masses at third points)

Massless beam, span l, equal masses m at l/3 and 2l/3. Flexibility coefficients: $a_{11} = a_{22} = 4l^3/243EI = 8/486\ (l^3/EI)$, $a_{12} = a_{21} = 7l^3/486EI$.

$[D] = m(l^3/486EI)[\ 8\ 7\ ;\ 7\ 8\ ]$; $\lambda = 1/\omega^2 = (m\ l^3/486EI)(8 \pm 7) = 15,\ 1\ (\times\ m\ l^3/486EI)$

$$\omega_1 = \sqrt{(486/15)}\ \sqrt{(EI/ml^3)} = 5.69\ \sqrt{(EI/ml^3)}\ \text{mode}\ (1,\ 1)$$
$$\omega_2 = \sqrt{486}\ \sqrt{(EI/ml^3)} = 22.05\ \sqrt{(EI/ml^3)}\ \text{mode}\ (1,\ -1) \quad (7a)$$

Bounds on fundamental frequency (3-mass chain, equal m, k)

Rayleigh (trial shape X = (1, 2, 3)): $\{X\}^T[K]\{X\} = 3k$, $\{X\}^T[M]\{X\} = 14m \rightarrow \omega_1^2 \approx 0.2143$ k/m (upper bound).

Dunkerley: $1/\omega_1^2 \approx \Sigma\ a_{ii}m_i = (1 + 2 + 3)\ m/k = 6m/k \rightarrow \omega_1^2 \approx 0.1667$ k/m (lower bound).

Exact: 0.1981 k/m, so $0.1667 < 0.1981 < 0.2143$ - the two simple methods bracket the true value.

4. Lumped Mass and Distributed Mass Systems

- Lumped (discrete) mass system: mass concentrated at finite points joined by massless elastic elements. Finite DOF, matrix equations (ODEs).
- Distributed (continuous) system: mass & elasticity spread continuously (beams, shafts, strings, rods). Infinite DOF; governed by partial differential equations; infinite natural frequencies.
- Lumping: a continuous system is approximated by lumped masses at nodes (finite-element idea). Accuracy improves with more masses; lowest few frequencies are good, higher ones poor.

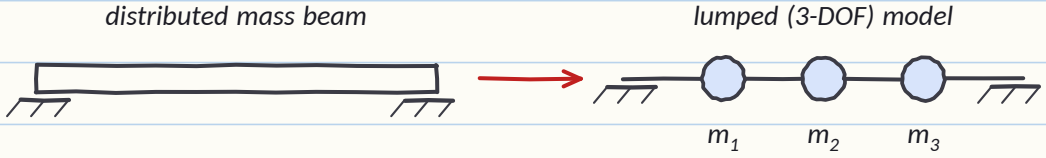


Fig 4.3 Lumping

Continuous member	Natural frequencies ω_n (rad/s)
String (tension P, mass/length ρA)	$(n\pi/l)\sqrt{(P/\rho A)}$ n = 1, 2, 3...
Bar, longitudinal; fixed-free	$(2n - 1)(\pi/2l)\sqrt{(E/\rho)}$
Bar, longitudinal; fixed-fixed or free-free	$(n\pi/l)\sqrt{(E/\rho)}$
Shaft, torsional; fixed-free	$(2n - 1)(\pi/2l)\sqrt{(G/\rho)}$
Beam, simply supported	$(n\pi/l)^2\sqrt{(EI/\rho A)}$
Beam, cantilever	$(\beta_n l)^2/l^2\sqrt{(EI/\rho A)}$; $\beta_1 l = 1.875$, $\beta_2 l = 4.694$, $\beta_3 l =$ 7.855
Beam, fixed-fixed	same form, $\beta_1 l = 4.730$, $\beta_2 l = 7.853$, $\beta_3 l = 10.996$

Rayleigh's quotient (fundamental frequency estimate)

Assume a mode shape y_i (static deflection curve is excellent). From $T_{max} = U_{max}$:

$$\omega_1^2 \approx g \Sigma W_i y_i / \Sigma W_i y_i^2 = \Sigma m_i g y_i / \Sigma m_i y_i^2 \; (= \{X\}^T[K]\{X\} / \{X\}^T[M]\{X\}) \quad (8)$$

- Always gives ω_1 higher than (or equal to) the true value (upper bound); Dunkerley (Unit I) gives lower bound - the true value lies between.

5. Stodola Method (Matrix Iteration)

Iterative method that converges to the fundamental mode (lowest ω) first, using the dynamic matrix $[D] = [a][M]$: $[D]\{X\} = (1/\omega^2)\{X\}$.

Procedure (fundamental mode)

1. Form $[D] = [a][M]$ (flexibility matrix × mass matrix).
2. Assume a trial mode $\{X\}_1$ (e.g. all 1's, or the expected shape).
3. Compute $\{X\}_2' = [D]\{X\}_1$. Normalise (divide by its first or largest element) to get $\{X\}_2$.
4. Repeat: $\{X\}_{n+1}' = [D]\{X\}_n$. The normalising factor converges to $\lambda_1 = 1/\omega_1^2$, and $\{X\}_n$ to the first mode shape.
5. Stop when successive λ and mode vectors agree to required accuracy. $\omega_1 = 1/\sqrt{\lambda_1}$.

- Higher modes: remove the lower mode by sweeping (orthogonality) $[D]_2 = [D]([I] - \{X\}_1\{X\}_1^T[M] / M_1)$, $M_1 = \{X\}_1^T[M]\{X\}_1$, and iterate again → ω_2, etc. (Round-off error spoils accuracy as higher modes are reached.)
- Why it works: any trial vector = $\Sigma c_i\{X\}^{(i)}$; each multiplication by $[D]$ multiplies

mode i by $\lambda_i = 1/\omega_i^2$; the largest λ (lowest ω) soon dominates.

Problem 2 (Stodola on the 3-mass chain)

Equal m, k (Fig 4.1, fixed-free). Flexibility matrix of the chain: $a_{ij} = \min(i,j)/k$. So $[D] = (m/k)[\ 1\ 1\ 1\ ;\ 1\ 2\ 2\ ;\ 1\ 2\ 3\ ]$. Let $\rho = m/k$. Find ω_1.

Iteration	Trial {X}	[D]{X}/ρ	Normalised	Factor
1	1, 1, 1	3, 5, 6	1, 1.667, 2.000	3.000
2	1, 1.667, 2.000	4.667, 8.333, 10.333	1, 1.786, 2.214	4.667
3	1, 1.786, 2.214	5.000, 9.000, 11.214	1, 1.800, 2.243	5.000
4	1, 1.800, 2.243	5.043, 9.086, 11.329	1, 1.802, 2.246	5.043
exact			1, 1.802, 2.247	5.049

$\lambda_1 = 5.049\ m/k \rightarrow \omega_1^2 = 0.1981\ k/m$, $\omega_1 = 0.445\ \sqrt{(k/m)}$ (matches exact $2(1 - \cos \pi/7)$).

For $m = 10\ kg$, $k = 4000\ N/m$: $\omega_1 = 0.445 \times 20 = 8.90\ rad/s$.

Exact frequencies of this chain: $\omega^2 = (k/m)[0.198, 1.555, 3.247] \rightarrow \omega = (0.445, 1.247, 1.802)\sqrt{(k/m)} = 8.90, 24.94, 36.04\ rad/s$; modes (1, 1.802, 2.247), (1, 0.445, -0.802), (1, -1.247, 0.555).

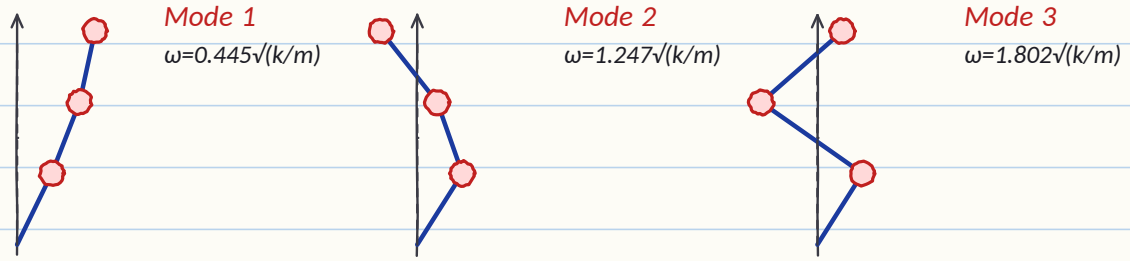
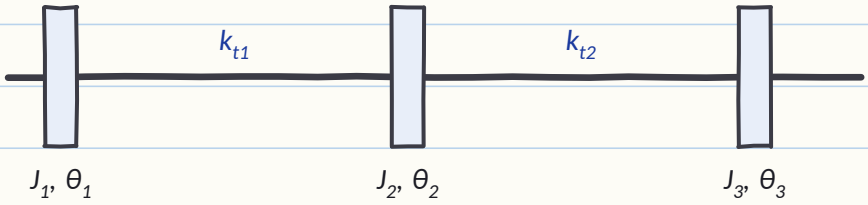


Fig 4.4 Mode shapes of the 3-DOF chain (amplitudes sideways)

6. Holzer's Method

Tabular trial-and-error method for natural frequencies and mode shapes of torsional (also spring-mass) systems with many rotors, in chain form (also geared systems after reduction). Gives any mode, not just the lowest.



free-free: $T_n = 0$ at end

(fixed end: $\theta_n = 0$)

Fig 4.5 Torsional chain for Holzer's method

Principle

At a natural frequency the free system satisfies both end conditions. Free end: zero torque. Assume ω^2, start from one end with $\theta_1 = 1$ and propagate:

$$\text{Inertia torque of rotor } i: T_i = \omega^2 J_i \theta_i$$
$$\text{Shaft torque beyond rotor } i: S_i = \Sigma_{j \le i}\, \omega^2 J_j \theta_j$$
$$\text{Twist of shaft } i: \theta_{i+1} = \theta_i - S_i / k_{t,i} \quad (9)$$

At the last rotor compute residual torque $R(\omega^2) = \Sigma_{all}\, \omega^2 J_j \theta_j$ (free end must be 0). Plot R against ω^2; zeros of R are the natural frequencies (rigid-body mode $\omega = 0$ for free-free).

Steps

1. Choose trial ω^2; $\theta_1 = 1$.
2. Fill table columns: J, θ, $\omega^2 J\theta$, $\Sigma\omega^2 J\theta$ (torque), torque/k_t, next θ.
3. Residual R at end. Repeat for several ω^2; sign change → root nearby; interpolate / refine until $R \approx 0$.
4. The θ values at the root = mode shape. For a fixed end the check is $\theta_n = 0$ instead of $R = 0$.

Problem 3 (three equal rotors, free-free)

$J_1 = J_2 = J_3 = 1$ kg·m^2, $k_{t1} = k_{t2} = 1$ N·m/rad. Find natural frequencies by Holzer.

ω^2	θ_1	Σ after 1	θ_2	Σ after 2	θ_3	Residual
0.5	1	0.500	0.500	0.750	-0.250	0.625
1.0	1	1.000	0.000	1.000	-1.000	0.000
2.0	1	2.000	-1.000	0.000	-1.000	-2.000
3.0	1	3.000	-2.000	-3.000	1.000	0.000

Residual changes sign between 0.5 and 2 and is zero at $\omega^2 = 1$ and 3. Hence $\omega = 0$ (rigid), $\omega_1^2 = 1 \rightarrow \omega = 1$ rad/s, $\omega_2^2 = 3 \rightarrow \omega = 1.732$ rad/s.

Mode shapes (θ_1, θ_2, θ_3): $\omega^2 = 1 \rightarrow (1, 0, -1)$ (node at middle rotor); $\omega^2 = 3 \rightarrow (1, -2,$

1) (two nodes).

- Matches exact eigenvalues 0, 1, 3 of [1 -1 0 ; -1 2 -1 ; 0 -1 1]. For real shafts, k_t = GJ/l and J_i = rotor inertias; geared branches are reduced to one shaft first.

Problem 3a (Holzer, fixed-free spring-mass chain)

Chain of Fig 4.1, m = 1, k = 1 (wall - k - m_1 - k - m_2 - k - m_3). Start at the free end: $x_3 = 1$; $F = \Sigma\, \omega^2 m\, x$; $x_{i-1} = x_i - F/k$. Wall condition: $x_0 = 0$.

ω^2	x_3	x_2	x_1	F at wall	x_0 (residual)
0.1	1	0.900	0.710	0.261	+0.449
0.1981	1	0.802	0.445	0.445	≈ 0
0.3	1	0.700	0.190	0.567	-0.377

Sign change of x_0 between 0.1 and 0.3 → root near 0.198: $\omega_1^2 = 0.1981\ k/m$; mode $(x_1, x_2, x_3) = (0.445, 0.802, 1) \sim (1, 1.802, 2.247)$. Repeating with trial ω^2 near 1.5 and 3.2 gives the higher modes.

7. Modal Analysis - Free and Forced Vibration

Uses orthogonality of modes to convert n coupled equations into n independent SDOF equations. Mode vectors $\{\varphi\}_i$:

$$\{\varphi\}_i^T[M]\{\varphi\}_j = 0,\ \{\varphi\}_i^T[K]\{\varphi\}_j = 0\ (i \neq j)$$
$$\{\varphi\}_i^T[M]\{\varphi\}_i = M_i\ (\text{modal mass}),\ \{\varphi\}_i^T[K]\{\varphi\}_i = K_i = \omega_i^2 M_i \qquad (10)$$

Mass-normalised modes ($M_i = 1$): $[\Phi]^T[M][\Phi] = [I]$, $[\Phi]^T[K][\Phi] = \text{diag}(\omega_i^2)$.

Modal transformation

Put $\{x\} = [\Phi]\{q\}$ (q = principal / modal co-ordinates), pre-multiply by $[\Phi]^T$:

$$M_i \ddot{q}_i + C_i \dot{q}_i + K_i q_i = Q_i(t) = \{\varphi\}_i^T\{F(t)\}\ i = 1\ldots n$$
$$\text{or } \ddot{q}_i + 2\zeta_i\omega_i \dot{q}_i + \omega_i^2 q_i = Q_i/M_i \qquad (11)$$

- Damping decouples only if it is proportional (Rayleigh): $[C] = \alpha[M] + \beta[K] \rightarrow \zeta_i = \tfrac{1}{2}(\alpha/\omega_i + \beta\omega_i)$.
- Each $q_i(t)$ is a SDOF solution (Units I, II); final response $\{x\} = \Sigma\, \{\varphi\}_i\, q_i(t)$ - often only the first few modes are needed.

Free vibration (initial conditions)

Initial modal values: $q_i(0) = \{\varphi\}_i^T[M]\{x_0\} / M_i$, $\dot{q}_i(0) = \{\varphi\}_i^T[M]\{\dot{x}_0\} / M_i$.

$$q_i(t) = q_i(0) \cos \omega_i t + (\dot{q}_i(0)/\omega_i) \sin \omega_i t \quad \{x\} = \Sigma \{\varphi\}_i \, q_i(t) \tag{12}$$

Forced (harmonic) vibration

For $F = \{F_0\} \sin \omega t$: $Q_i = \{\varphi\}_i^T\{F_0\} \sin \omega t$, and (undamped):

$$q_i = (Q_{i0}/M_i) / (\omega_i^2 - \omega^2) \sin \omega t \quad \{x\} = \Sigma \{\varphi\}_i \, q_i \tag{13}$$

- Mode i is strongly excited near $\omega = \omega_i$; a mode is not excited if $\{\varphi\}_i^T\{F_0\} = 0$ (force applied at a node, or orthogonal to the mode).

Problem 4 (modal analysis of a 2-DOF system)

$m_1 = m_2 = 10$ kg, $k_1 = k_2 = k_3 = 4000$ N/m. (a) Free response if $x_0 = (1, 0)$ mm, at rest. (b) Forced response for $F = 20 \sin 10t$ N on m_1.

Modes: $\{\varphi\}_1 = (1, 1)$, $\omega_1^2 = k/m = 400$; $\{\varphi\}_2 = (1, -1)$, $\omega_2^2 = 3k/m = 1200$. $M_1 = M_2 = 20$ kg.

(a) $q_1(0) = \{\varphi\}_1^T M \, x_0/M_1 = 10(1)/20 = 0.5$; $q_2(0) = 0.5 \rightarrow$

$x_1 = 0.5 \cos 20t + 0.5 \cos 34.64t$, $x_2 = 0.5 \cos 20t - 0.5 \cos 34.64t$ (mm) (beats between modes).

(b) $Q_{10} = (1, 1)\cdot(20, 0) = 20$; $Q_{20} = 20$. $q_1 = (20/20)/(400 - 100) = 3.333\times10^{-3}$ m; $q_2 = 1/(1200 - 100) = 0.909\times10^{-3}$ m

$X_1 = q_1 + q_2 = 4.24$ mm, $X_2 = q_1 - q_2 = 2.42$ mm (agrees with direct solution $X_1 = 7000(20)/33\times10^6$).

Problem 5 (orthogonality check, 3-mass chain)

Modes: $\{\varphi\}_1 = (1, 1.802, 2.247)$, $\{\varphi\}_2 = (1, 0.445, -0.802)$, $m = 1$.

$\{\varphi\}_1^T[M]\{\varphi\}_2 = 1 + 1.802(0.445) + 2.247(-0.802) = 1 + 0.802 - 1.802 = 0$ (orthogonal).

Modal masses: $M_1 = 1 + 3.247 + 5.049 = 9.296$ m; $M_2 = 1 + 0.198 + 0.643 = 1.841$ m. Mass-normalised: $\{\varphi\}_1/\sqrt{9.296} = (0.328, 0.591, 0.737)$.

Truncation of modes

- Response is dominated by modes whose frequencies are near the excitation and whose modal force $\{\varphi\}_i^T\{F\}$ is large; high modes respond quasi-statically.
- Mode-acceleration / truncation: keep first few modes (say $p < n$) → big saving for large FE models.

Comparison of approximate methods

Method	Gives	Remark
Rayleigh	ω_1	upper bound; needs a good trial shape
Dunkerley	ω_1	lower bound; needs only a_{ii}
Stodola	ω_1, then higher	iterative; sweeping for higher modes
Holzer	any ω, modes	tabular; chain / torsional systems
Modal analysis	response	exact superposition after eigen-solution

Problem 6 (Lagrange with damping and force)

Masses m_1, m_2; springs k_1 (wall - m_1), k_2 (m_1 - m_2); dashpot c between m_2 and ground; force $F(t)$ on m_2.

$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2$; $U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2$; $D = \tfrac{1}{2}c\,\dot{x}_2^2$; $Q_1 = 0$, $Q_2 = F(t)$.

$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0$$
$$m_2\ddot{x}_2 + c\,\dot{x}_2 - k_2x_1 + k_2x_2 = F(t) \qquad (13a)$$

Problem 7 (Rayleigh proportional damping)

For the chain with $\omega_1 = 8.90$, $\omega_2 = 24.94$ rad/s, require $\zeta_1 = 0.02$, $\zeta_2 = 0.03$ in $[C] = \alpha[M] + \beta[K]$, $\zeta_i = \tfrac{1}{2}(\alpha/\omega_i + \beta\omega_i)$.

$0.04 = \alpha/8.90 + 8.90\beta$; $0.06 = \alpha/24.94 + 24.94\beta \rightarrow \beta = 2.10\times10^{-3}$ s, $\alpha = 0.1896$ s^{-1}

Then ζ_3 ($\omega_3 = 36.04$) = $\tfrac{1}{2}(0.1896/36.04 + 0.0021\times36.04) = 0.041$ - higher modes are damped more.

Problem 8 (modal forced response, 3-mass chain)

$m = 10$ kg, $k = 4000$ N/m each; $F = 10 \sin 15t$ N on m_3. Find amplitudes.

$\omega_i^2 = 400 \times (0.1981, 1.555, 3.247) = 79.2, 622.0, 1298.8$; modes (1, 1.802, 2.247), (1, 0.445, -0.802), (1, -1.247, 0.555)

Modal masses $M_i = 10\{\varphi\}^T\{\varphi\} = 92.96, 18.41, 28.63$ kg; modal forces $Q_{i0} = \varphi_i(3)F_0 = 22.47, -8.02, 5.55$ N

$q_i = (Q_{i0}/M_i)/(\omega_i^2 - 225) = -1.658, -1.097, +0.181$ (mm)

$X_1 = \Sigma\varphi_i(1)q_i = -2.58$ mm, $X_2 = -3.70$ mm, $X_3 = -2.75$ mm (negative sign: out of phase with force, since 15 rad/s lies above $\omega_1 = 8.9$).

Check row 1 of equations: $(8000 - 10\times225)(-2.575) - 4000(-3.701) \approx 0$.

8. Quick Revision & Short Questions

Method	Key idea / formula
Lagrange	$d/dt(\partial T/\partial \dot{q}) - \partial T/\partial q + \partial D/\partial \dot{q} + \partial U/\partial q = Q$
Influence coefficients	$\{F\} = [K]\{x\}$, $\{x\} = [a]\{F\}$, $[a] = [K]^{-1}$; $[D] = [a][M]$
Rayleigh quotient	$\omega^2 \approx \{X\}^T[K]\{X\} / \{X\}^T[M]\{X\}$ (upper bound)
Stodola	$\{X\}_{n+1} = [D]\{X\}_n$; factor $\to 1/\omega_1^2$; sweep for higher modes
Holzer	$\theta_{i+1} = \theta_i - \Sigma\omega^2 J\theta/k_t$; residual = 0 at natural frequency
Modal analysis	$\{x\} = [\Phi]\{q\}$; $M_i\ddot{q}_i + K_i q_i = \{\varphi\}_i^T\{F\}$

1. **Why use generalised co-ordinates?** They are independent, automatically satisfy constraints and make Lagrange's equation simple.
2. **State Maxwell's reciprocal theorem.** $a_{ij} = a_{ji}$: deflection at i due to unit load at j equals that at j due to unit load at i.
3. **Which mode does Stodola give first and why?** The fundamental, since [D] has the largest eigenvalue $1/\omega_1^2$.
4. **When does the Holzer residual vanish?** Only at natural frequencies (boundary condition at last rotor satisfied).
5. **Why is modal analysis efficient?** n coupled equations become n SDOF equations; few modes suffice for response.
6. **Lumped vs distributed system?** Finite vs infinite DOF; ODEs vs PDEs.
7. **What is Rayleigh damping?** $[C] = \alpha[M] + \beta[K]$; diagonalised by the undamped modes, so modal equations stay uncoupled.
8. **What is sweeping in Stodola's method?** Using orthogonality to remove an already-found mode from [D] so iteration converges to the next mode.
9. **Rayleigh vs Dunkerley?** Rayleigh gives an upper bound, Dunkerley a lower bound for ω_1.
10. **Number of natural frequencies of a continuous beam?** Infinite (governed by a PDE); lumping retains only n of them.

Applications

- Turbine/generator shaft trains (Holzer), multi-storey buildings (shear-building models), aircraft wing and fuselage (modal testing), machine-tool structures, automobile body-in-white (FE modal analysis).

Formula sheet - Unit IV

- Eigenproblem: $[K]\{X\} = \omega^2[M]\{X\}$; $|[K] - \omega^2[M]| = 0$; $[D] = [K]^{-1}[M] = [a][M]$, $\lambda = 1/\omega^2$.
- Lagrange: $d/dt(\partial T/\partial \dot{q}_i) - \partial T/\partial q_i + \partial D/\partial \dot{q}_i + \partial U/\partial q_i = Q_i$.
- Chain, n equal masses, fixed-free: $\omega_i^2 = 2(k/m)[1 - \cos((2i - 1)\pi/(2n + 1))]$.
- Rayleigh: $\omega^2 \approx \{X\}^T[K]\{X\}/\{X\}^T[M]\{X\}$; Dunkerley: $1/\omega_1^2 \approx \Sigma\, a_{ii}m_i$.
- Holzer: $\theta_{i+1} = \theta_i - (\Sigma_{j \le i}\, \omega^2 J_j\theta_j)/k_{t,i}$; residual = 0 at natural frequency.
- Modal: $\{x\} = [\Phi]\{q\}$; $M_i\ddot{q}_i + C_i\dot{q}_i + K_iq_i = \{\varphi\}_i^T\{F\}$; $q_i(0) = \{\varphi\}_i^T[M]\{x_0\}/M_i$.

Exam tips

- For 2-DOF and 3-DOF: always show frequency equation, roots, mode ratios and a neat sketch of mode shapes with nodes.
- In Stodola/Holzer problems present the iteration/table clearly and state convergence criterion and final ω with units.
- Quote Maxwell's theorem whenever [a] or [K] is shown symmetric; use consistent units (N, m, kg, rad/s).

Important questions: (1) Lagrange's equation applied to a 2/3-DOF system (2) influence coefficients and Maxwell's theorem (3) Stodola iteration problem (4) Holzer table (5) modal analysis of free and forced vibration.